

KICC National Exhibition Platform

Technical Achievements Report

August 2 - August 20, 2026 | Confidential

Executive Summary

The KICC National Exhibition Platform is a production-grade digital infrastructure spanning all 47 Kenyan counties. In 18 days, the engineering team delivered a complete ecosystem: Laravel application serving kicctest.org through Cloudflare edge network, TiDB Cloud production database, R2 media CDN, marketplace commerce, trade system, 47-county admin infrastructure, and an AI-powered 3D/immersive content pipeline. This report documents the technical architecture, capabilities, and integration status of every system component.

1. Platform Architecture

The platform operates as a multi-tier system with four distinct layers:

Edge Layer

Cloudflare
worker with r

Application Layer

Laravel 1
Mother Admin

Database Layer

TiDB Cloud
connections f

Media & Pipeline Layer

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2. County System (47 Counties)

All 47 Kenyan counties are fully onboarded with individual public websites, sector-specific content, tourism attractions, hotels, product listings, and trade data. Each county has a full-featured professional admin dashboard for content management, image uploads, pricing, advertising, 4D immersive video management, sector toggling, entity management, and CSV report generation. Counties are organized by region (Central, Coast, Eastern, Nairobi, North Eastern, Nyanza, Rift Valley, Western) with searchable, filterable directory.

3. Commerce & Marketplace

Full marketplace with product listings, variants, pricing, stock tracking, and category associations. Orders processed through escrow system with buyer/seller protection. Supports search, filtering, booking flows, and complete order management with payment status tracking.

4. Trade System

Comprehensive trade module connecting Kenyan counties to international markets across trading blocs: EAC, COMESA, AfCFTA, and WTO. Export eligibility checker determines tariff preferences. Export enquiry workflow creates trade lead records integrated with n8n automation. 10 trade agreements seeded across 4 trading blocs with full public-facing pages.

5. Admin Control Panels

KICC Mother Admin: central dashboard overseeing all 47 counties, national government exhibitors, private exhibitors, marketplace orders, provider certifications, escrow transactions, users, and landing page hero media. Includes Artisan command console and provider certification queue.

47 County Professional Admin Dashboards: each manages content, images, prices, advertising, 4D immersive videos, sectors, entities, and CSV reports.

County Exhibitor Portal: trade board members track products, exhibitors, orders, and escrow revenue.

6. Media Asset Pipeline

Centralized MediaAsset system managing every image, video, and 3D model in Cloudflare R2 storage, served via CDN with CORS headers, range request support, and immutable cache. Supports: video posters, thumbnails, WebP conversion, HLS streaming, and 4D immersive video slots per sector entity. No hardcoded paths - everything resolves through the database at runtime.

7. AI & 3D Content Pipeline

OpenRouter AI gateway for 200+ models. Custom 3D reconstruction pipeline: photogrammetry (SfM), Gaussian splatting, depth estimation, Blender walkthrough rendering. Multi-agent orchestration system with actor/decider/observer/learner architecture. 81-second cinematic hero video generated from building photographs with TV-news overlays, Ken Burns effects, and professional color grading.

8. Infrastructure & Security

Cloudflare Workers edge gateway with caching, rate limiting, and JWT validation. Nginx + PHP-FPM configured for 1GB upload support. GitHub Actions CI/CD pipeline. Laravel Horizon for queue management, Pulse for performance monitoring, Sentry for error tracking. Email verification, Google OAuth, MFA/TOTP, role-based access control (kicc_admin, county_admin, exhibitor, provider), Sanctum API tokens, Passport OAuth2. 148 automated tests with 202 assertions.

9. Integration Status

Integration pipelines built for 27 services across 5 categories. 9 government APIs (KRA, Huduma, NTSA, KWS, Tourism, Museums, Health, Agriculture, Education) have complete service classes and graceful fallback - awaiting API keys to go live. 6 payment drivers (M-Pesa STK Push, Stripe, Airtel Money, T-Kash, Bank EFT, Crypto) fully coded in stub mode. 6 media pipeline engines (Wan2GP, Hailuo, Kling, Wan, Tripo3D, ROAD) disabled pending configuration. 3 N8n production workflows built but undeployed. 26 platform events wired for webhook automation.